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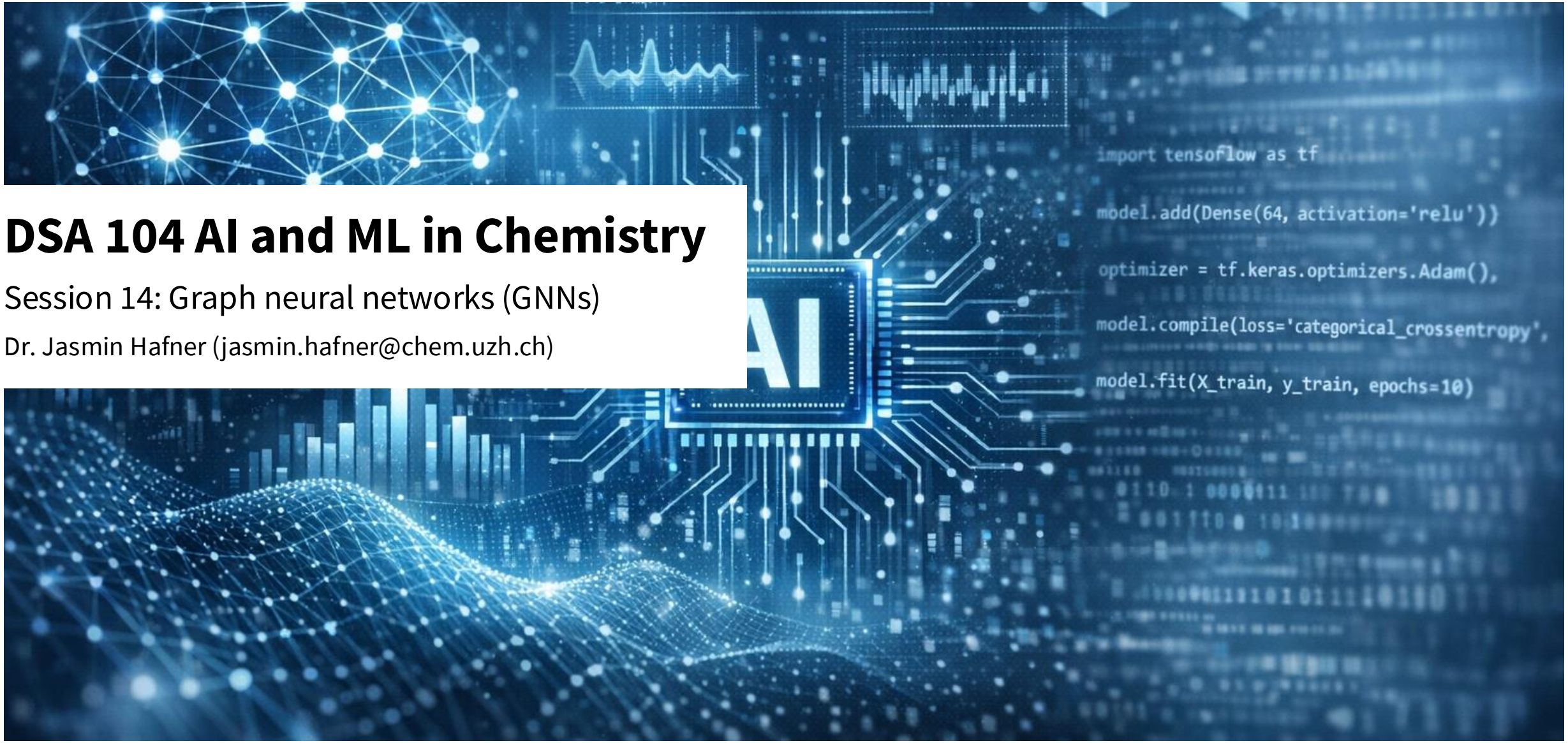
Department of Chemistry



DSA 104 AI and ML in Chemistry

Session 14: Graph neural networks (GNNs)

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Lesson outline

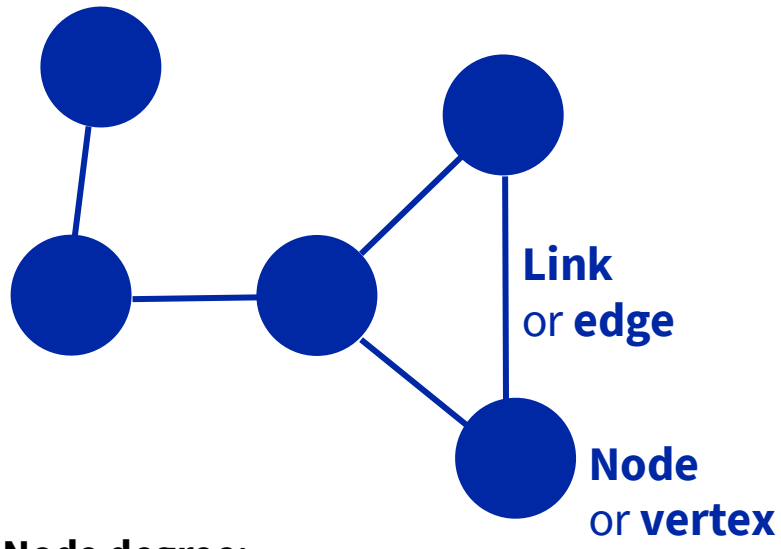
Recap: Graph theory

Architecture of Graph Neural Networks (GNNs)

Applications in chemistry

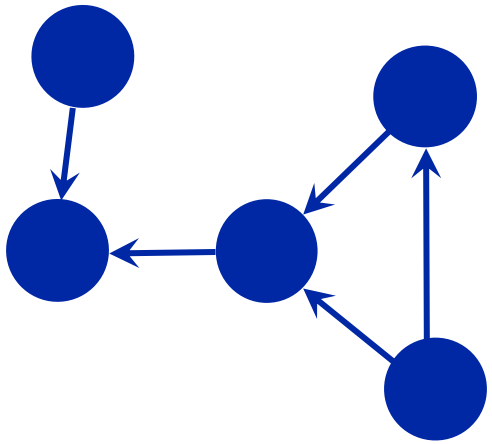
Recap: graph theory

Simple, undirected graph

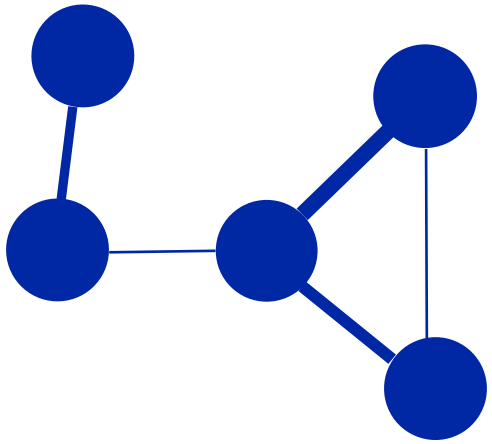


Node degree:
number of connections

Directed graph



Weighted graph

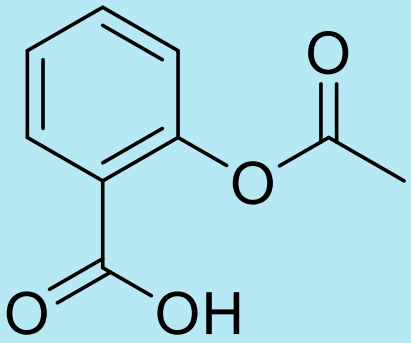


https://www.geeksforgeeks.org/deep-learning/what-are-graph-neural-networks/

Discussion

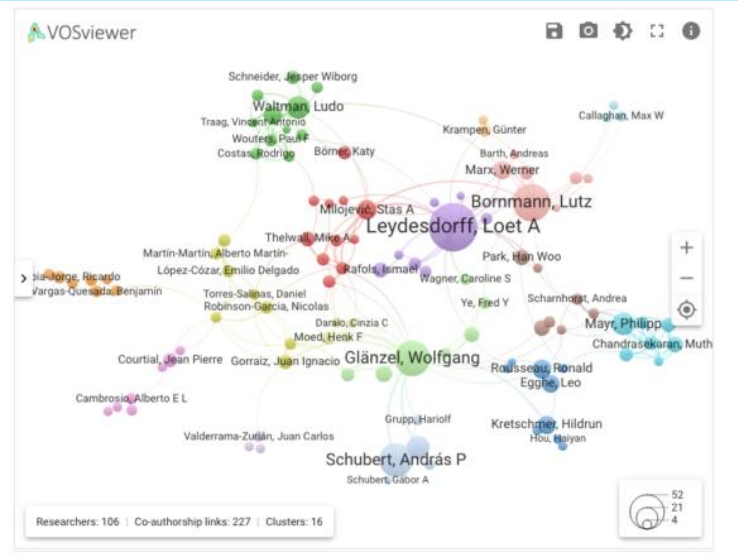
- What kind of data can be represented by graphs?

Molecular graph



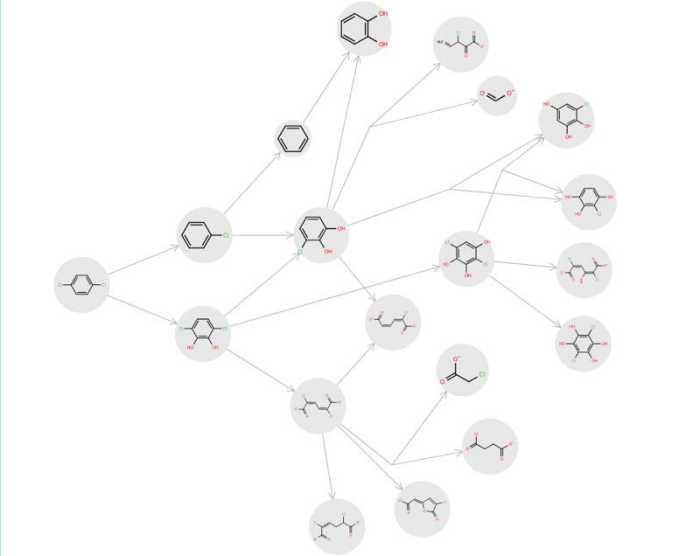
Nodes: atoms
Edges: bonds

Citation network



Nodes: Articles
Edges: Citations

Reaction network

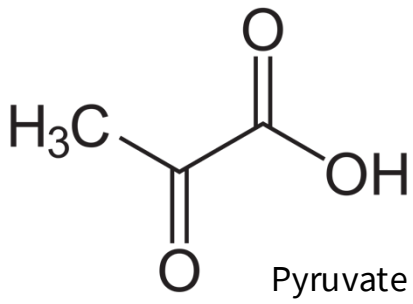


Nodes: compounds
Edges: reactions

https://vosviewer.com, https://legacy.envipath.org

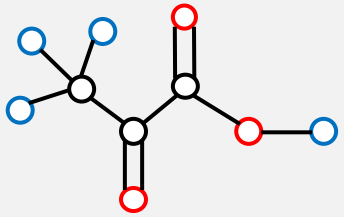
Molecular graphs

How is a molecule represented in a computer (e.g., *inside* RDKit) ?



Pyruvate

Graph representation
atoms = nodes
bonds = edges



Matrix
Bond-electron matrix:

	C	C	O	C	O	O
C	0	1	0	0	0	0
C	1	0	2	1	0	0
O	0	2	0	0	0	0
C	0	1	0	0	2	1
O	0	0	0	2	0	0
O	0	0	0	1	0	0

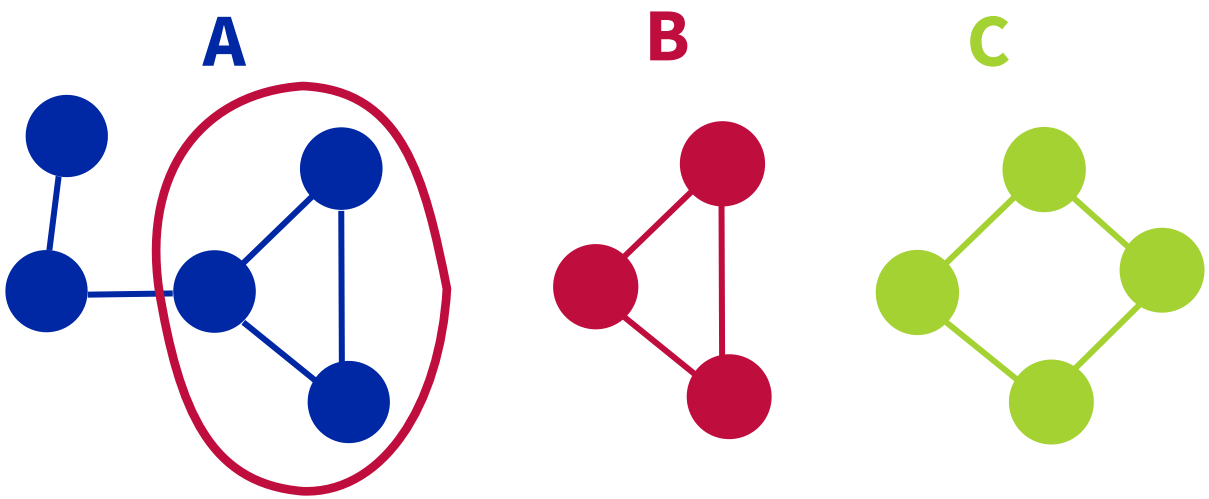
String
SMILES representation:

CC(=O)C(=O)O

Molecular graphs in cheminformatics

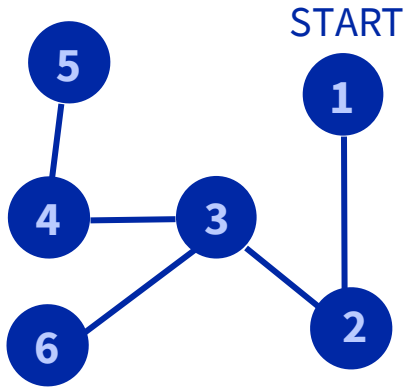
(Sub)graph matching:

Is **B** in **A** ?
Is **C** in **A**?



- Search for **functional** groups in molecule
- Search database for structures containing specific **molecular moiety**

Graph traversal (e.g., to build SMILES):



Represent graph as a path (e.g., for SMILES)
Depth-first: 1-2-3-4-5-6
Breadth-first: 1-2-3-4-6-5
…

RDKit stores molecules as graphs: https://deepwiki.com/rdkit/rdkit/2.1-molecular-data-structures

Exercise: Molecular graphs

1) Create your own Graph class

2) Save a (simple) molecule of your choice as a Graph object

3) Traverse the molecule and print a linear path (similar to a SMILES code)

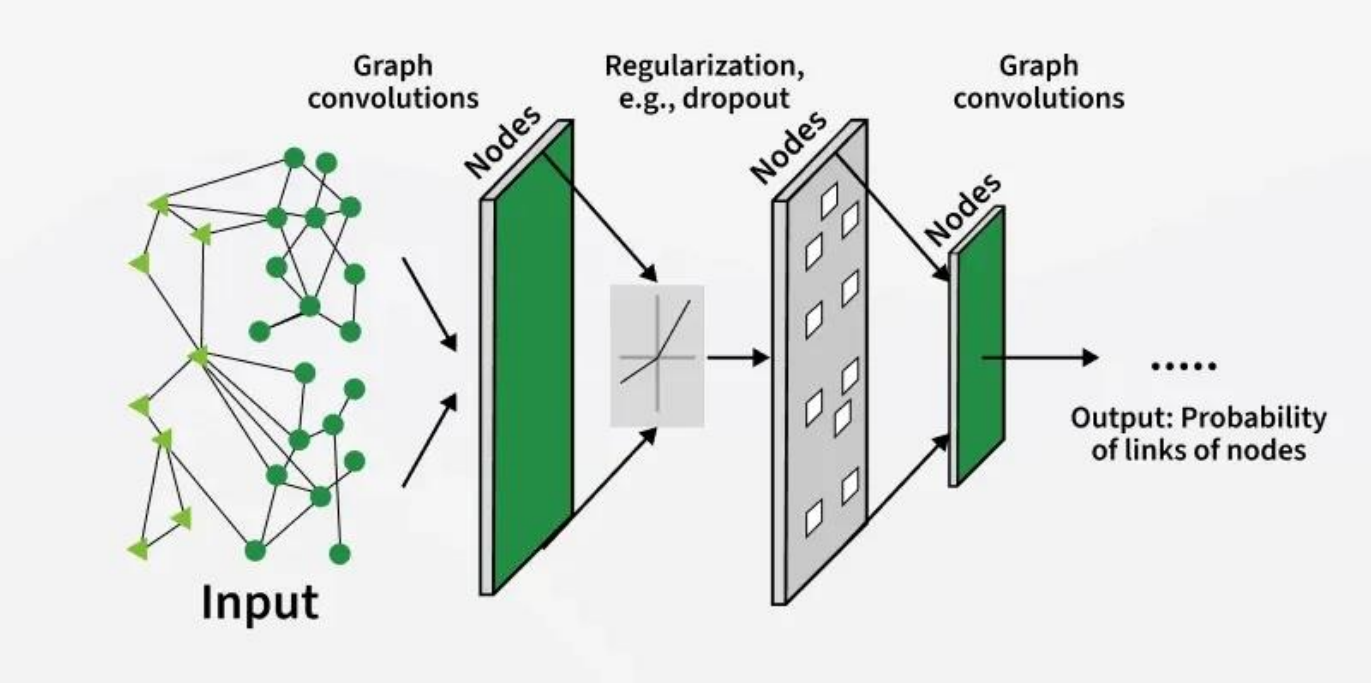
4) You want to find whether your molecule has an aldehyde/ketone group. How do you do this?

5) Can you define a function that reduces the aldehyde/ketone to an alcohol group?

Graph neural networks (GNNs)

GNNs capture relationships, dependencies and interactions between entities:

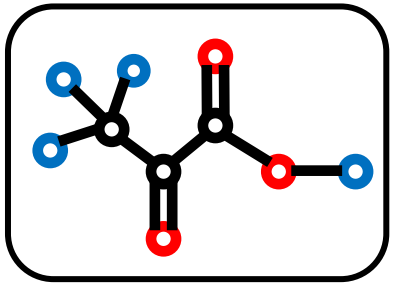
- Operate on **graph** objects
- Information is passed between connected nodes (neighbors) through **message-passing** steps.
- **Permutation invariance**: Changing the order of nodes in the graph representation (e.g., in adjacency matrix) should not affect outcome
- Typical applications:
  - social network analysis
  - **molecule prediction**
  - recommendation systems



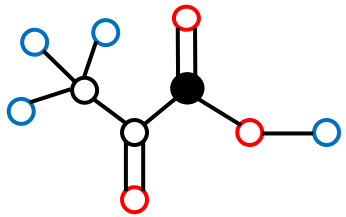
https://www.geeksforgeeks.org/deep-learning/what-are-graph-neural-networks/

Applications of GNNs representing molecules

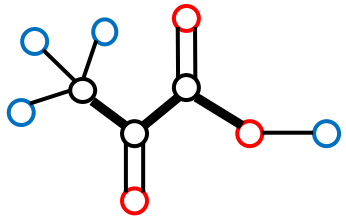
1. Graph-level tasks:
   - predict a **property** of the entire graph/molecule (e.g., toxicity)
   - predict entirely **new graphs/molecules** (e.g., drug discovery)



2. Node-level tasks:
   - predict a **node property** of a specific atom (e.g., atomic charges)
   - predict a **new node to be added** to the graph.



3. Edge-level tasks:
   - predict **edge properties** (e.g., intramolecular forces between atoms)
   - predict **new edge** (bonds) in the graph



https://projects.volkamerlab.org/teachopencadd/talktorials/T035_graph_neural_networks.html

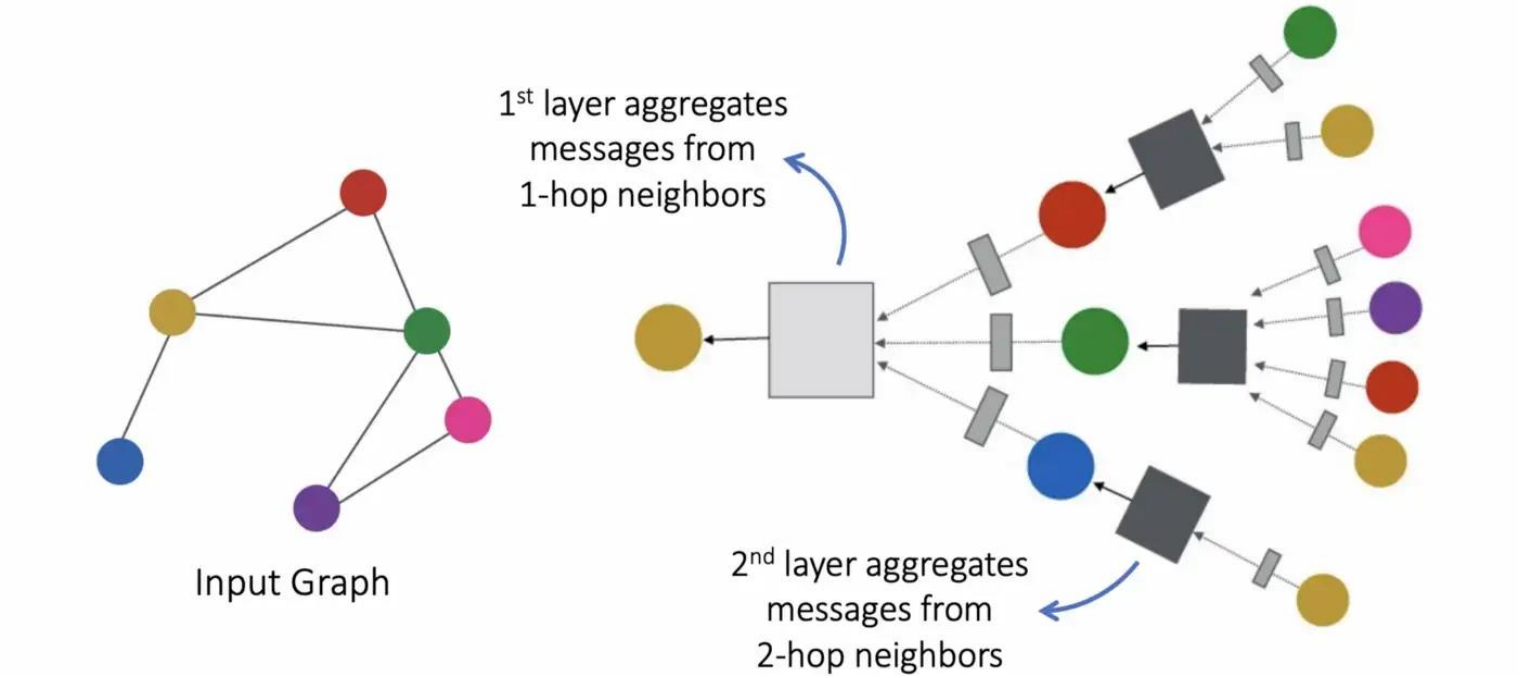
Message passing in GNNs

- GNNs have **message-passing layers** where information about neighboring nodes is collected

- For each node:
  - look at the direct **neighbors** and gather **information**
  - **aggregate** all information about neighbors (e.g., mean, sum)
  - **update node** with aggregated information

- Each node contains the information about the **direct neighbors (1-hop)**.

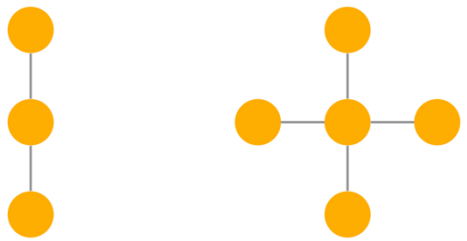
- Repeat this n times, we aggregate information about **the n closest neighbors (n-hop)**.



https://projects.volkamerlab.org/teachopencadd/talktorials/T035_graph_neural_networks.html

Types of GNNs used

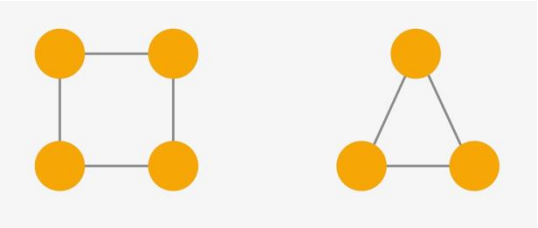
1. **Graph Convolutional Networks (GCN)**
   - Uses the **mean** to aggregate node and edge properties
     - Pros: simple
     - Cons: Different graphs can lead to same embedding



Indistinguishable by GCN if node properties are equal

2. **Graph Isomorphism Networks (GIN)**
   - Uses the **sum** to aggregate node and edge properties
   - Pros: still simple, better differentiates between non-isomorphic graphs

     → **suited to represent molecules**



Indistinguishable by GIN if node properties are equal

3. **Graph Attention Networks (GAT)**
   - self-**attention** mechanism applied directly to the **graph** structure
   - Pros: **interpretation** possible

https://projects.volkamerlab.org/teachopencadd/talktorials/T035_graph_neural_networks.html

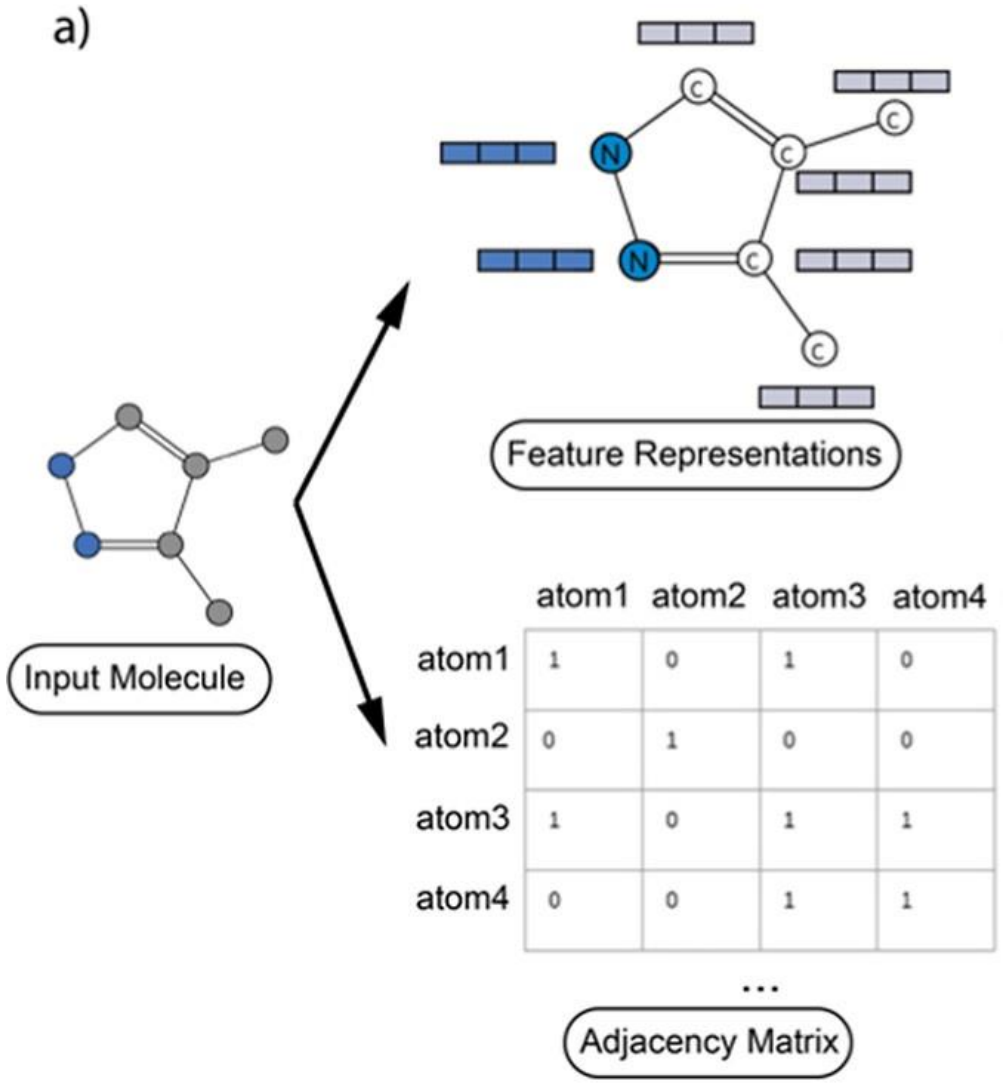
Featurization: describes the properties of nodes and edges

Node features:

- atom type
- formal charge
- hybridization type
- hydrogen bonding
- aromaticity
- degree
- number of hydrogens
- chirality
- and partial charge
- …

Edge features:

- bond type
- ring
- conjugation
- stereo configuration
- …



K. Zhang, P. Schwaller and K. Fenner, *Environ. Sci. Technol.*, 2025, **59**, 4518–4529.

Exercise: GNN

Scenario: Lipophilicity/hydrophobicity (logP, octanol-water partition coefficient) is an important proxy for numerous more complex molecular properties (e.g., toxicity, Koc) . We try to train a GNN that can predict logP using the ESOL data set.

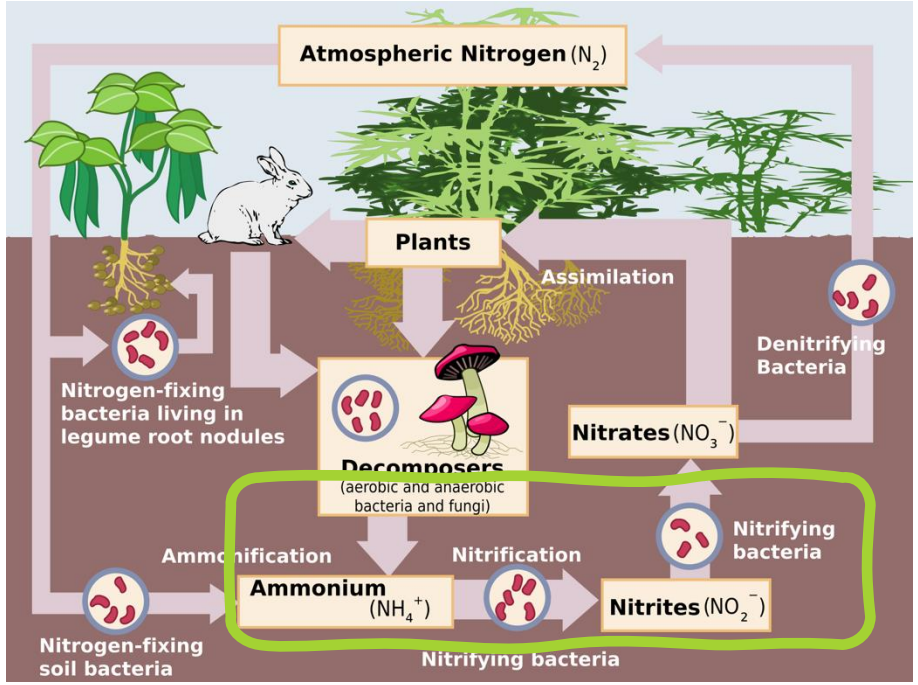
Test GCN, GIN and GAT: Which one works better to predict logP from structure?

Application example: GAT for environmental toxicity prediction

- Objective: predict **toxicity** of chemicals on **microbial nitrification** by **ammonia-oxidizing bacteria** (AOB)

- Challenge: small experimental data set

- Approach: Graph Attention Network (GAT)
  - **Pretraining** on **big** dataset (14'000 chemicals, logP) (→ upcoming lecture)
    - → **logP** is a good predictor of **baseline toxicity**
    - → **Assumption**: microbial nitrification may suffer from membrane disturbance by baseline toxicants
  - **Fine-tuning** on **small** dataset for nitrification inhibition (288 chemicals)

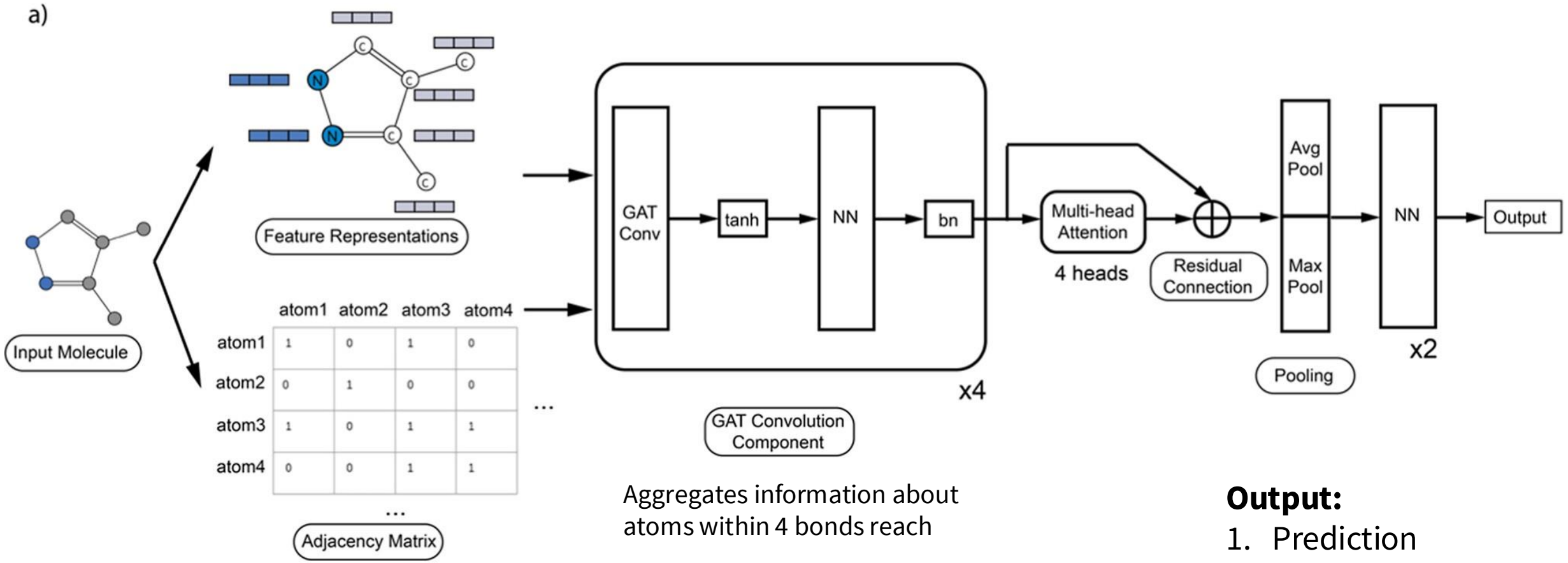


- Implementation: sequential Graph Attention Networks Convolution (GATConv) layers using PyTorch

- Interpretability: use **attention scores** do detect chemical substructures indicating toxicity ("structural alerts")

K. Zhang, P. Schwaller and K. Fenner, *Environ. Sci. Technol.*, 2025, **59**, 4518–4529., Image: https://en.wikipedia.org/wiki/Nitrification#/

Architecture



Aggregates information about atoms within 4 bonds reach

Output:

1. Prediction
2. Attention

Implementation: https://github.com/zhangky12/ToxPred_nitrification/blob/main/model.py

K. Zhang, P. Schwaller and K. Fenner, *Environ. Sci. Technol.*, 2025, **59**, 4518–4529.

Prediction performance

Table 1. Test Performance of Different Models[a]

no	model	mean ROC-AUC	standard deviation
1	SVM	0.79	0.10
2	RF	0.83	0.09
3	AdaBoost	0.80	0.10
4	XGBoost	0.83	0.07
5	KNN	0.80	0.12
6	MLP	0.81	0.09
7	ChemBERTa	0.79	0.05
8	MolFormer	**0.85**	0.07
9	MolCLR_GIN	0.71	0.09
10	GAT	0.82	0.11
11	MolCLR_GAT	0.83	0.10
12	FTGAT	**0.85**	0.08

Supervised learning using **MACCS**

Self-supervised/ **pretrained** on millions of molecules and **finetuned** on toxicity data

→ Not pretrained

→ **Pretrained** GAT on MolCLR data & finetuned

→ **Pretrained** with **LogP** data & finetuned

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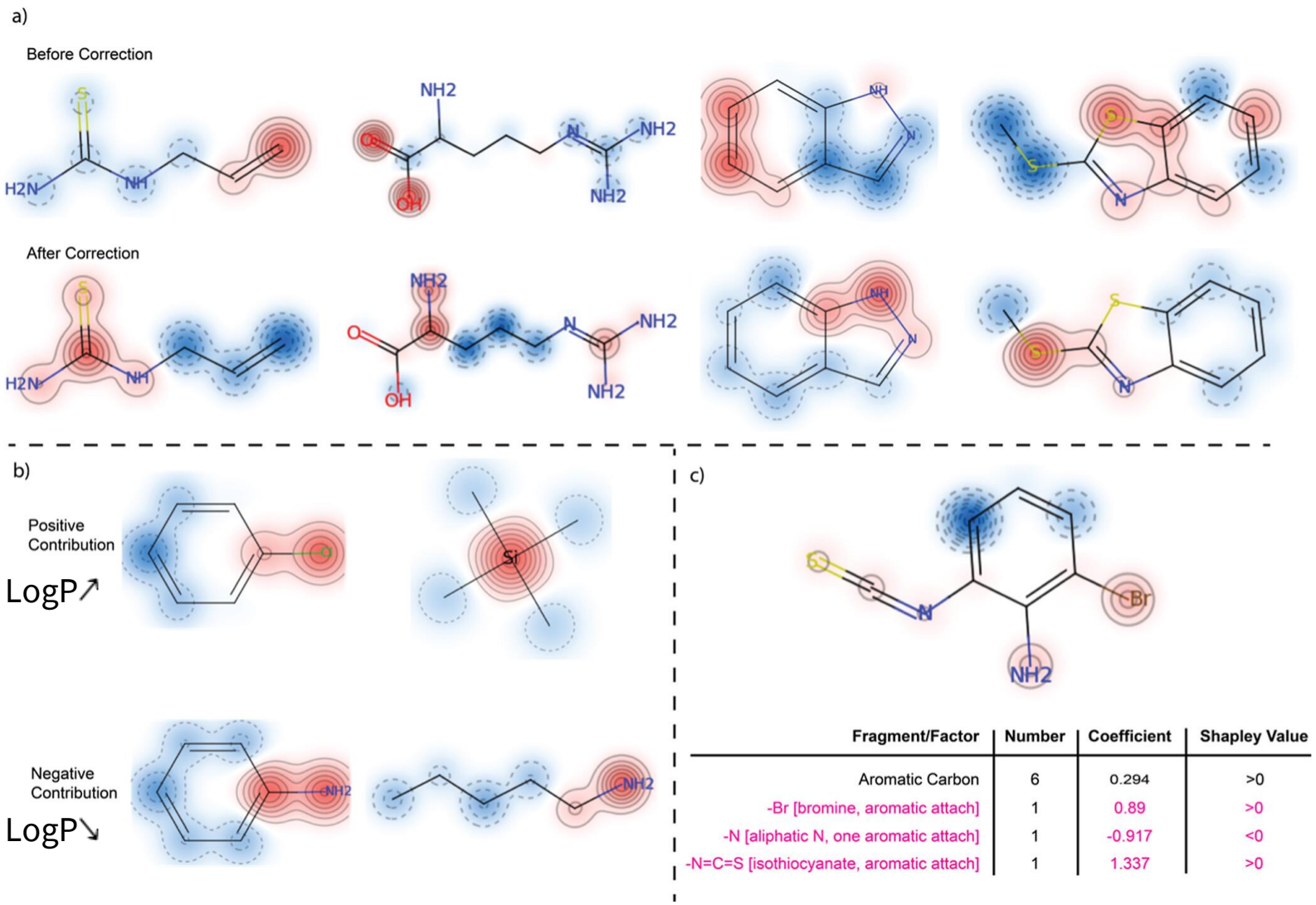
Model interpretation: attention mechanisms

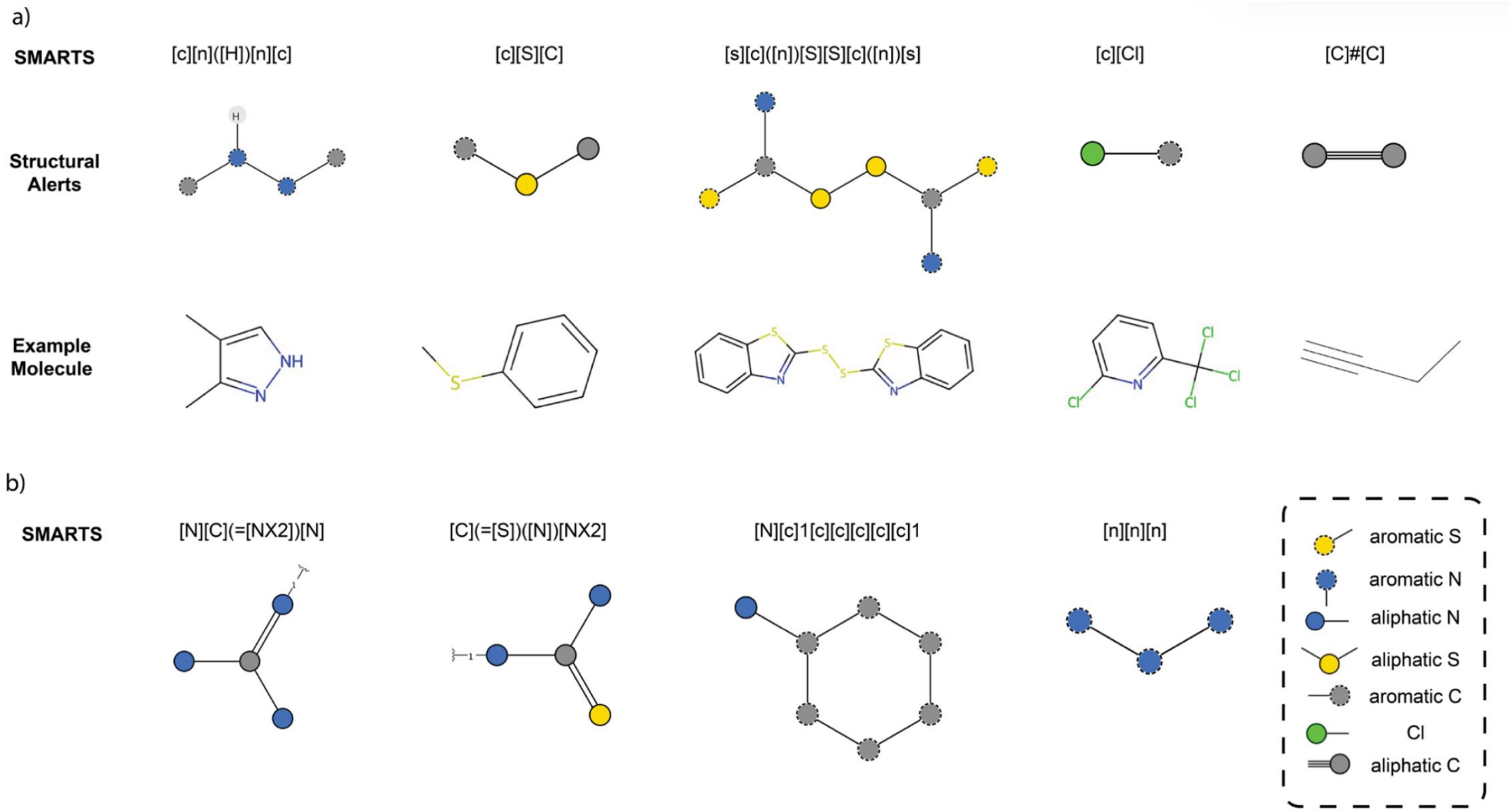


Fragment/Factor	Number	Coefficient	Shapley Value
Aromatic Carbon	6	0.294	>0
-Br [bromine, aromatic attach]	1	0.89	>0
-N [aliphatic N, one aromatic attach]	1	-0.917	<0
-N=C=S [isothiocyanate, aromatic attach]	1	1.337	>0

Low attention
High attention

K. Zhang, P. Schwaller and K. Fenner, *Environ. Sci. Technol.*, 2025, **59**, 4518–4529.

Outcome: Structural alerts for environmental toxicity prediction



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